Day 5 : FIRST_VALUE & LAST_VALUE

Questions with Answers

Tables

Table 1 : employees

emp_name	dept	salary
Amit	IT	60000
Neha	IT	75000
Ravi	IT	90000
Pooja	HR	40000
Karan	HR	45000
Simran	HR	52000
Rahul	Sales	35000
Anjali	Sales	48000

Table 2 : stock_trades

stock	trade_time	price
TCS	09:15:00	3500
TCS	10:30:00	3520
TCS	12:00:00	3490
TCS	15:30:00	3550
INFY	09:15:00	1500
INFY	11:00:00	1480
INFY	13:45:00	1510
INFY	15:30:00	1525

Questions & Answers

Q1.

Show every employee along with the **lowest salary in the whole company**.

Table used: employees

Answer

emp_name	dept	salary	lowest_salary
Rahul	Sales	35000	**35000**
Pooja	HR	40000	**35000**
Karan	HR	45000	**35000**
Anjali	Sales	48000	**35000**
Simran	HR	52000	**35000**
Amit	IT	60000	**35000**
Neha	IT	75000	**35000**
Ravi	IT	90000	**35000**

Q2.

Show every employee along with the **highest salary in the whole company** using LAST_VALUE **without** a frame clause. What do you notice? Why does it happen?

Table used: employees

Answer

emp_name	dept	salary	highest_salary
Rahul	Sales	35000	35000
Pooja	HR	40000	40000
Karan	HR	45000	45000
Anjali	Sales	48000	48000
Simran	HR	52000	52000
Amit	IT	60000	60000
Neha	IT	75000	75000
Ravi	IT	90000	90000

Wrong result! Every row just shows its own salary. With ORDER BY, the default frame is "first row up to the CURRENT row", so the last row of each window is the row itself.

Q3.

Fix Q2: show every employee along with the **highest salary in the company** using LAST_VALUE correctly.

Table used: employees

Answer

emp_name	dept	salary	highest_salary
Rahul	Sales	35000	**90000**
Pooja	HR	40000	**90000**
Karan	HR	45000	**90000**
Anjali	Sales	48000	**90000**
Simran	HR	52000	**90000**
Amit	IT	60000	**90000**
Neha	IT	75000	**90000**
Ravi	IT	90000	**90000**

With the full frame (all rows), the last row is truly the highest salary: 90000.

Q4.

For each employee, show the **lowest** and **highest salary of their department**, and the **gap** between the department's highest salary and their salary.

Table used: employees

Answer

emp_name	dept	salary	dept_lowest	dept_highest	gap_from_top
Pooja	HR	40000	**40000**	**52000**	**12000**
Karan	HR	45000	**40000**	**52000**	**7000**
Simran	HR	52000	**40000**	**52000**	**0**
Amit	IT	60000	**60000**	**90000**	**30000**
Neha	IT	75000	**60000**	**90000**	**15000**
Ravi	IT	90000	**60000**	**90000**	**0**
Rahul	Sales	35000	**35000**	**48000**	**13000**
Anjali	Sales	48000	**35000**	**48000**	**0**

Q5.

Show the **name of the top earner** of each department next to every employee.

Table used: employees

Answer

emp_name	dept	salary	top_earner
Simran	HR	52000	**Simran**
Karan	HR	45000	**Simran**
Pooja	HR	40000	**Simran**
Ravi	IT	90000	**Ravi**
Neha	IT	75000	**Ravi**
Amit	IT	60000	**Ravi**
Anjali	Sales	48000	**Anjali**
Rahul	Sales	35000	**Anjali**

Q6.

For every trade, show the stock's **opening price** (first trade of the day) and **closing price** (last trade of the day).

Table used: stock_trades

Answer

stock	trade_time	price	open_price	close_price
INFY	09:15:00	1500	**1500**	**1525**
INFY	11:00:00	1480	**1500**	**1525**
INFY	13:45:00	1510	**1500**	**1525**
INFY	15:30:00	1525	**1500**	**1525**
TCS	09:15:00	3500	**3500**	**3550**
TCS	10:30:00	3520	**3500**	**3550**
TCS	12:00:00	3490	**3500**	**3550**
TCS	15:30:00	3550	**3500**	**3550**

Q7.

Show **one row per stock** with its open price, close price and the **day's change** (close - open).

Table used: stock_trades

Answer

stock	open_price	close_price	day_change
INFY	1500	1525	**25**
TCS	3500	3550	**50**

Q8.

For every trade, show how much the price **moved compared with the stock's opening price** of the day.

Table used: stock_trades

Answer

stock	trade_time	price	move_from_open
INFY	09:15:00	1500	**0**
INFY	11:00:00	1480	**-20**
INFY	13:45:00	1510	**10**
INFY	15:30:00	1525	**25**
TCS	09:15:00	3500	**0**
TCS	10:30:00	3520	**20**
TCS	12:00:00	3490	**-10**
TCS	15:30:00	3550	**50**

Q9.

Show the **lowest-paid employee's name** in each department next to every employee.

Table used: employees

Answer

emp_name	dept	salary	lowest_paid
Pooja	HR	40000	**Pooja**
Karan	HR	45000	**Pooja**
Simran	HR	52000	**Pooja**
Amit	IT	60000	**Amit**
Neha	IT	75000	**Amit**
Ravi	IT	90000	**Amit**
Rahul	Sales	35000	**Rahul**
Anjali	Sales	48000	**Rahul**

Q10.

For each stock, show the **% change from open to close** (rounded to 2 decimals), one row per stock.

Table used: stock_trades

Answer

stock	pct_change
INFY	**1.67**
TCS	**1.43**

INFY: (1525 - 1500) x 100 / 1500 = 1.67% | TCS: (3550 - 3500) x 100 / 3500 = 1.43%